

Jadhav Ashvini Manohar

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Representations

- list1= characters list
- list2= integer list
- list3= mixed data type list

```
In [1]: list1=['w','e','l','c','o','m','e']  
list2=[10,20,30,40,50,60]  
list3=['w','e','l','c','o','m','e',10,20,30,40,50,60]
```

```
In [6]: for i in range(len(list1)):  
        if list1[i]=='e':  
            print(list1[i])
```

e
e

inbuilt functions

```
In [10]: list1=['w','e','l','c','o','m','e']  
list2=[10,20,30,40,50,60]  
list3=['w','e','l','c','o','m','e',10,20,30,40,50,60]
```

```
In [12]: print(type(list1))
```

<class 'list'>

```
In [14]: print(len(list1))
```

7

```
In [15]: print(max(list1))
```

w

```
In [16]: print(min(list1))
```

c

```
In [18]: print(sum(list2))
```

```
210
```

```
In [19]: print(type(list2))
```

```
<class 'list'>
```

```
In [20]: print(len(list2))
```

```
6
```

```
In [21]: print(max(list2))
```

```
60
```

```
In [22]: print(min(list2))
```

```
10
```

```
In [23]: print(type(list3))
```

```
<class 'list'>
```

```
In [24]: print(len(list3))
```

```
13
```

concatenation

```
In [26]: print(list1+list2)
```

```
['w', 'e', 'l', 'c', 'o', 'm', 'e', 10, 20, 30, 40, 50, 60]
```

```
In [31]: # with space  
print(list1+list3)
```

```
['w', 'e', 'l', 'c', 'o', 'm', 'e', 'w', 'e', 'l', 'c', 'o', 'm', 'e', 10, 20, 30, 40, 50, 60]
```

```
In [32]: print(list3+list2)
```

```
['w', 'e', 'l', 'c', 'o', 'm', 'e', 10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60]
```

repetition

```
In [33]: print(list2*3)
```

```
[10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60]
```

```
In [34]: print(list3*2)
```

```
['w', 'e', 'l', 'c', 'o', 'm', 'e', 10, 20, 30, 40, 50, 60, 'w', 'e', 'l', 'c', 'o', 'm', 'e', 10, 20, 30, 40, 50, 60]
```

```
In [35]: print(list1*4)
```

```
['w', 'e', 'l', 'c', 'o', 'm', 'e', 'w', 'e', 'l', 'c', 'o', 'm', 'e', 'w', 'e', 'l', 'c', 'o', 'm', 'e', 'w', 'e', 'l', 'c', 'o', 'm', 'e']
```

membership operator

```
In [36]: print('w' in list1)
```

True

```
In [37]: print('10' in list1)
```

False

```
In [38]: print('20' in list2)
```

False

```
In [39]: print('a' in list3)
```

False

index , slice

```
In [53]: print("First element :", list2[0])
print("Second element :", list2[1])
print("Third element :", list2[2])
print("Last element :", list2[5])
```

First element : 10
Second element : 20
Third element : 30
Last element : 60

```
In [50]: list1=['w','e','l','c','o','m','e']
for i in range(len(list1)):
    print(i,list1[i])
```

0 w
1 e
2 l
3 c
4 o
5 m
6 e

```
In [40]: list1[0]
```

Out[40]: 'w'

```
In [43]: list2[1:-6:-1]
```

Out[43]: [20]

```
In [45]: list3[1:15:2]
```

Out[45]: ['e', 'c', 'm', 10, 30, 50]

```
In [46]: list3[-12:-3]
```

Out[46]: ['e', 'l', 'c', 'o', 'm', 'e', 10, 20, 30]

```
In [47]: list1[:]
```

Out[47]: ['w', 'e', 'l', 'c', 'o', 'm', 'e']

```
In [48]: list3[:15]
```

```
Out[48]: ['w', 'e', 'l', 'c', 'o', 'm', 'e', 10, 20, 30, 40, 50, 60]
```

mutable vs immutable

```
In [54]: list1 = ['w','e','l','c','o','m','e']  
  
print("Before:", list1)  
  
list1[0] = 'W'  
  
print("After:", list1)
```

Before: ['w', 'e', 'l', 'c', 'o', 'm', 'e']

After: ['W', 'e', 'l', 'c', 'o', 'm', 'e']

```
In [55]: list2 = [10,20,30,40,50,60]  
  
print("Before:", list2)  
  
list2[2] = 300  
  
print("After:", list2)
```

Before: [10, 20, 30, 40, 50, 60]

After: [10, 20, 300, 40, 50, 60]

```
In [56]: list3 = ['w','e','l','c','o','m','e',10,20,30,40,50,60]  
  
print("Before:", list3)  
  
list3[7] = 100  
  
print("After:", list3)
```

Before: ['w', 'e', 'l', 'c', 'o', 'm', 'e', 10, 20, 30, 40, 50, 60]

After: ['w', 'e', 'l', 'c', 'o', 'm', 'e', 100, 20, 30, 40, 50, 60]

methods

```
In [60]: list1 = ['w','e','l','c','o','m','e']
```

```
In [62]: print(list1.count('e'))
```

2

```
In [64]: print(list1[0].upper())
```

E

```
In [68]: list1[0].lower()
```

```
Out[68]: 'e'
```

```
In [67]: list3=['w','e','l','c','o','m','e',10,20,30,40,50,60]
```

```
In [69]: print(list3[0].upper())  
print(list3[3].upper())
```

W
C

we can not use string methods to list2 because the elements are integers, not strings.